Model Calibration Under Label Drift

A. Rao, L. Chen, M. Iqbal

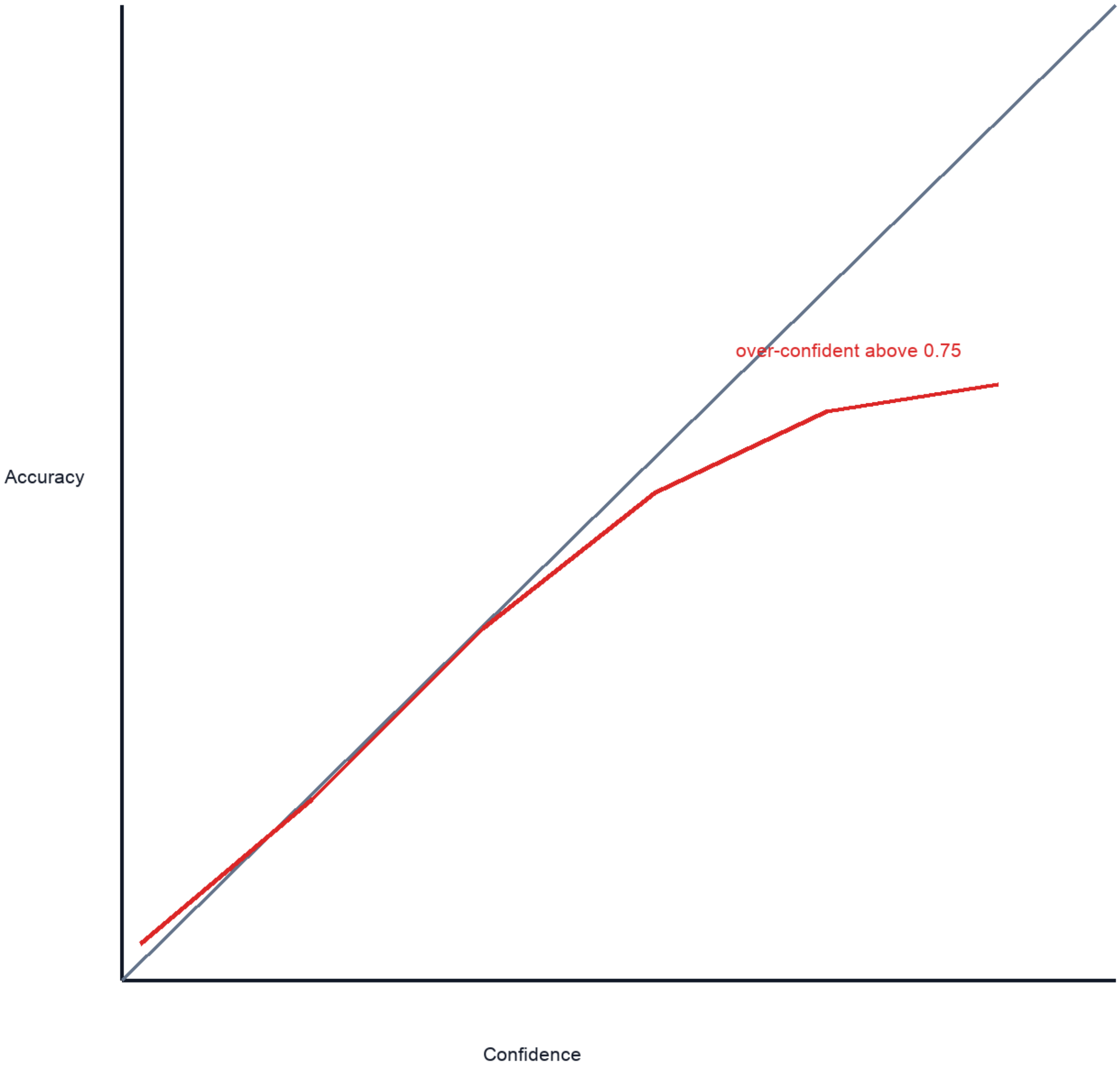
Abstract: We evaluate calibration under label drift using ten bins and a drift prior. The main result is that replay buffer improves F1 while lowering ECE.

ECE = sum_b (n_b / n) * |acc(b) - conf(b)|, B = 10

Method	ECE down	F1 up
Baseline	0.087	0.781
+ temperature scaling	0.041	0.779
+ drift prior	0.033	0.802
+ replay buffer	0.029	0.817

Sidebar note: Do not read this before the ablation table. The note explains why the replay buffer row is kept despite higher storage cost.

Figure 2 - Calibration Curves



Caption: dashed diagonal is perfect calibration. Model curve falls below diagonal above confidence 0.75.

Supplement Table S1 - continued

Rows continue on next page. Do not treat repeated header as data.

Cohort	Shift	n	ECE	F1	Note
A	none	1200	0.029	0.817	baseline final
B	mild	840	0.034	0.804	uses drift prior
C	moderate	610	0.052	0.781	see footnote 1

Footnote 1 begins: Cohort C excludes 17 records with missing confidence...

Supplement Table S1 - continued

Cohort	Shift	n	ECE	F1	Note
D	severe	455	0.071	0.744	fails threshold
E	recovered	500	0.038	0.799	replay restored
Total		3605			do not average ECE from this row

Footnote 1 continued: ...so reported n is after exclusion. Table S1 total row is a count total, not an ECE average.

Supplement Figure S2 - Drift Error Matrix

Color and letter both matter. A slash means the cohort/bin pair requires manual review.

Reviewer error level by confidence bin

	0.50	0.60	0.70	0.80	0.90
A none	L	L	M	M	H
B mild	L	M	M	H	H
C moderate	M	M	H	~~H~~	H
D severe	M	H	~~H~~	H	H
E recovered	L	~~M~~	M	H	M

Legend: L low error; M medium error; H high error. Slash means manual review required.

Caption: Cohort D has high error in every bin from 0.60 through 0.90. E recovered at 0.90 is medium, not high.